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OpenCV 5

What's New—and How to Prepare Your Company for the Upgrade

A practical guide for product teams

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DARPA Grand Challenge (2005)





Mars Helicopter Ingenuity’s first flight (2021)



OpenCV

Was used in both projects



OpenCV Library

An open-source library for computer vision and machine learning



GitHub stars

opencv/opencv
main repository



Downloads per week

opencv-python
package on PyPI



Years of open source

First public release
June 2000



Vision and ML algorithms

Optimized algorithms
in one library

Checked 7 October 2026. Sources: GitHub, PyPIStats, OpenCV.org

OpenCV at a glance

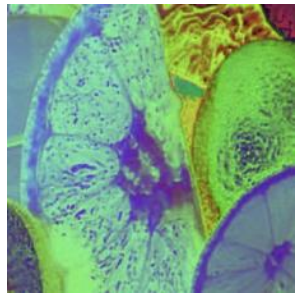


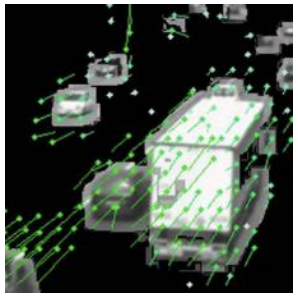
Image processing



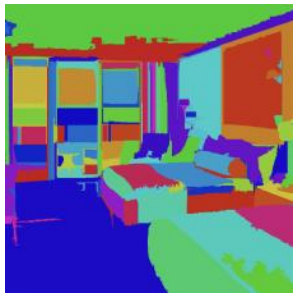
Geometric transforms



Shape fitting



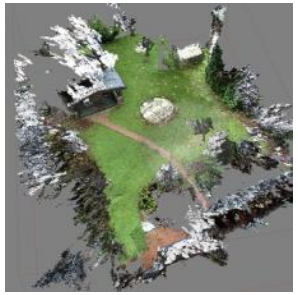
Optical flow
and tracking



Segmentation



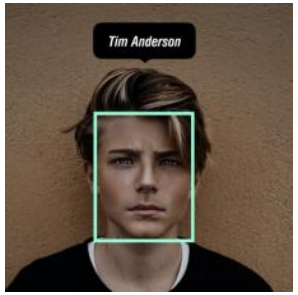
Camera calibration



Feature mapping



RGB-D and depth



Deep learning
and machine learning



Computational
photography

Shared foundation: data structures, numerical operations and image/video I/O

OpenCV milestones: 2000–2009

2000



Gary Bradski, OpenCV founder

Public debut at CVPR

Intel’s project reaches developers with its first public release.

2005



Stanley wins the DARPA Grand Challenge

Stanford’s autonomous vehicle used OpenCV in its vision system.

2009

A new C++ interface

OpenCV 2.0 introduces automatic memory management and a new API.

OpenCV milestones: 2015–2026

2015



OpenCV 3.0 expands the library

Contrib broadens the ecosystem. T-API adds OpenCL acceleration.

2018

OpenCV 4.0

OpenCV 4.0 released

C++11 foundation, G-API pipelines and ONNX model support.

2026

OpenCV 5.0 released

Released on 19 August 2026, with a modernized core, APIs and DNN engine.

Why companies use OpenCV

Build faster

Proven algorithms save development time.



Run efficiently

Optimized routines use hardware acceleration.

Stay in control

Open source you can inspect, modify, and ship.

Keep products running

Enterprise support for the releases your products depend on.

Your team owns the data, accuracy requirements and product validation.

What's New in OpenCV 5

1. Better language support
2. Faster & smaller core
3. New Hardware Acceleration Layer (HAL)
4. Cleaner API
5. Deep Neural Network (DNN) engine
6. Better 3D vision support
7. Better documentation



Core and developer experience

Language baseline

C++17

Python 3

Array shapes



Illustrative data shapes

Rebuild and compare behavior

Legacy C API and Python 2 removed. NumPy 2 support also exists in 4.x.

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OpenCV HAL

Hardware Acceleration

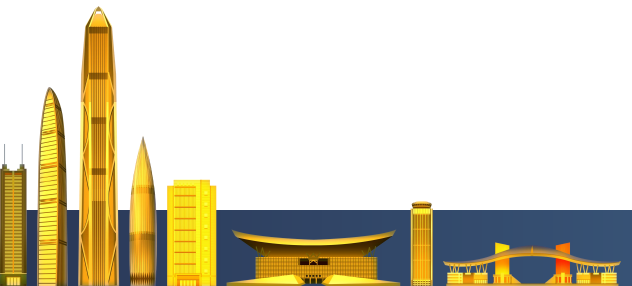


What is OpenCV HAL

Hardware Abstraction Layer (HAL) uses vendor provided libraries to accelerate OpenCV functions

The code remains the same

10% to over 300% speedup



Fast Operations with OpenCV HAL

Operation	OpenCV 4.13	OpenCV 5.0	Calculated speedup
Affine warp, linear	1.06 ms	0.85 ms	**1.25×**
Remap, linear	0.95 ms	0.70 ms	**1.36×**
Perspective warp, linear	1.62 ms	0.95 ms	**1.71×**
Transpose	0.15 ms	0.06 ms	**2.50×**
Laplacian, kernel size 3	0.83 ms	0.25 ms	**3.32×**

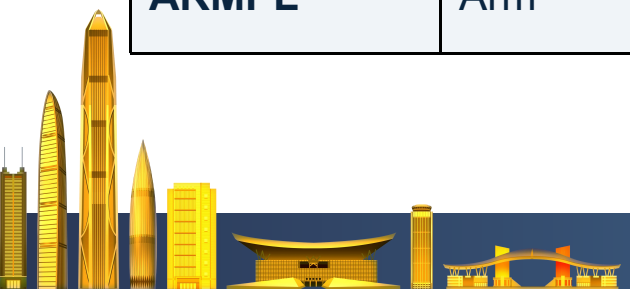
Selected pip-package results. HAL's contribution was not isolated.

The HAL ecosystem

Vendor backends can be enabled at build time with CMake.

BACKEND	VENDOR	TARGET	SCOPE	CMAKE FLAG
IPP/IPPICV	Intel	x86/x86-64	stats, transforms, warp	WITH_IPP
KleidiCV	Arm	AArch64 (NEON/SVE2/SME2)	color, filter, resize, warp	WITH_KLEIDICV
Carotene	NVIDIA	Arm NEON (v7/v8)	80+ imgproc/arith ops	WITH_CAROTENE
FastCV	Qualcomm	Snapdragon (Android/Linux Arm)	core + imgproc	WITH_FASTCV
RVV HAL	OpenCV China, SpacemiT	RISC-V RVV 1.0	~119 core+imgproc ops	WITH_HAL_RVV
NDSRVP	Andes	RISC-V P-extension (DSP)	filter, morph, warp, remap	WITH_NDSRVP
ARMPL	Arm	AArch64	BLAS/LAPACK linear algebra	WITH_ARMPL

Source: OpenCV 5 HAL documentation



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OpenCV DNN

Runs all modern neural networks. Often faster than ONNX.

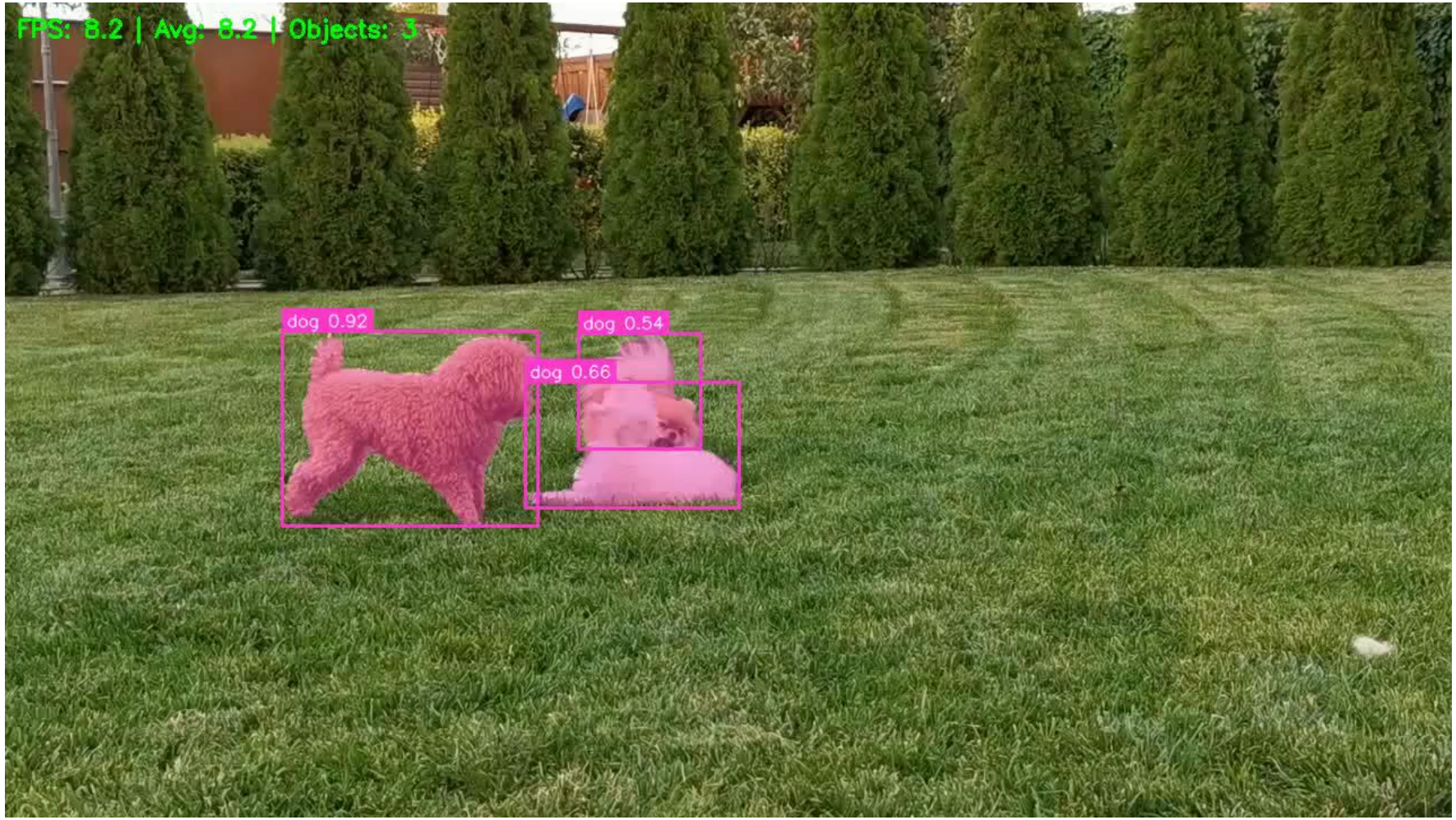


OpenCV 5 DNN : YOLO26 (Detection)



40% Faster than ONNX Runtime

OpenCV 5 DNN : RF-DETR (Segmentation)



RF-DETR Python demo from LearnOpenCV.

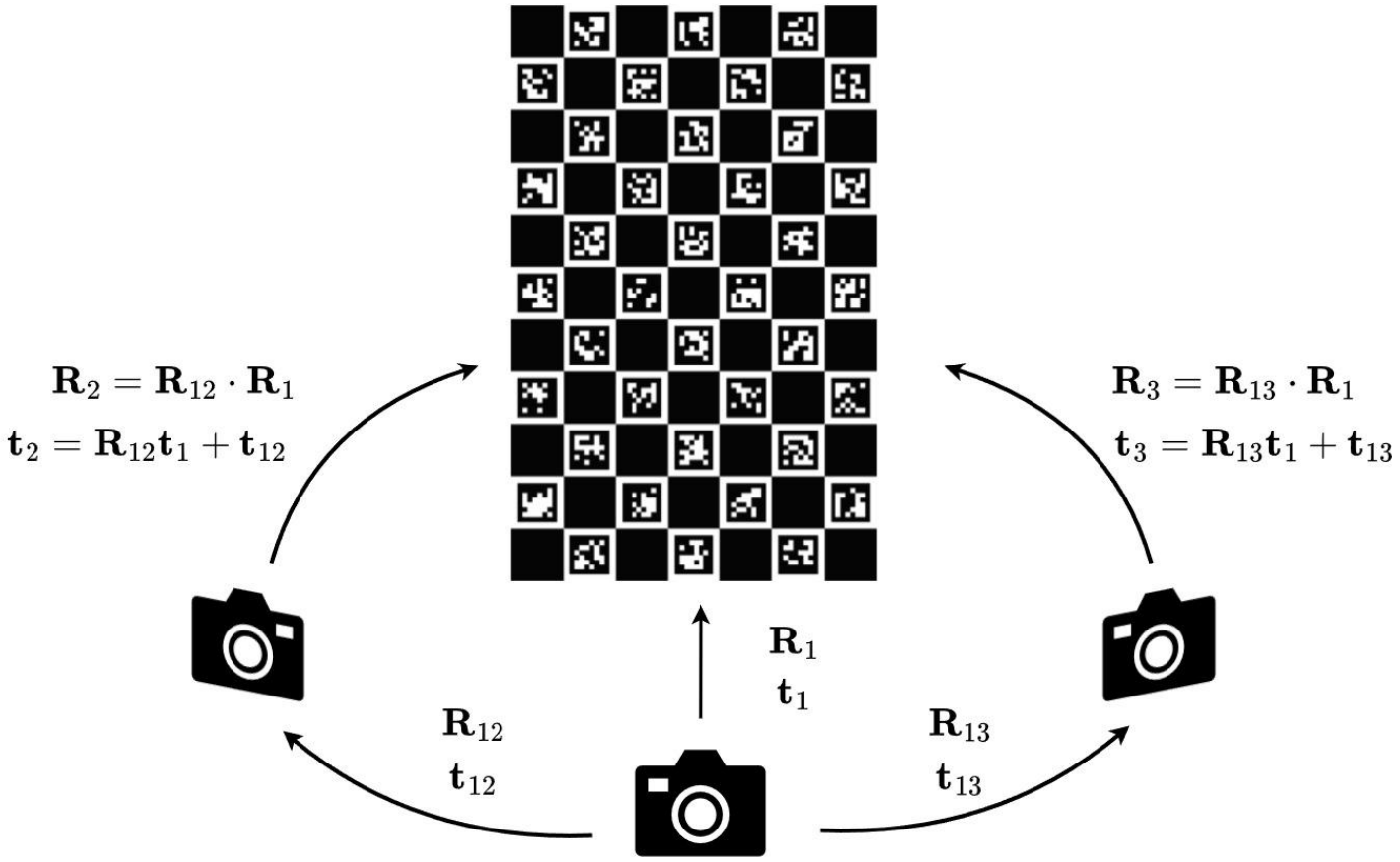
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Calibration



Multi-camera calibration



One calibration pipeline for the whole rig

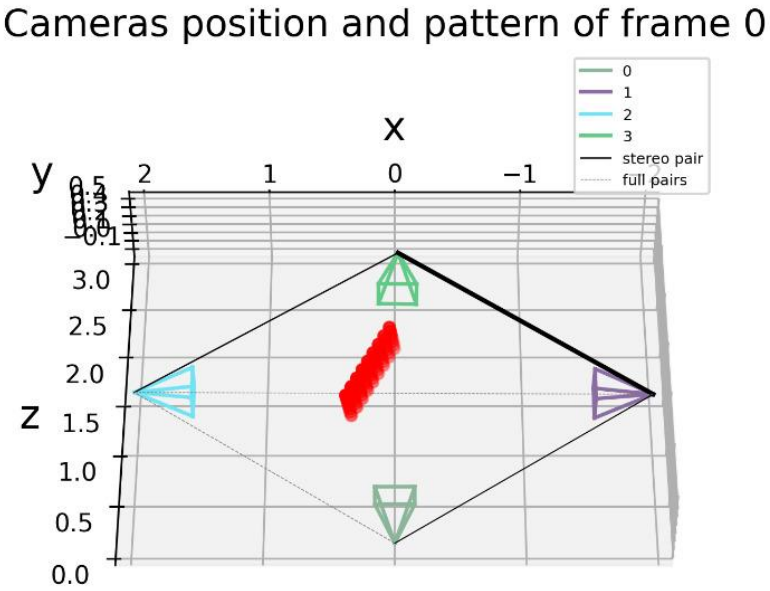
Pinhole + fisheye cameras

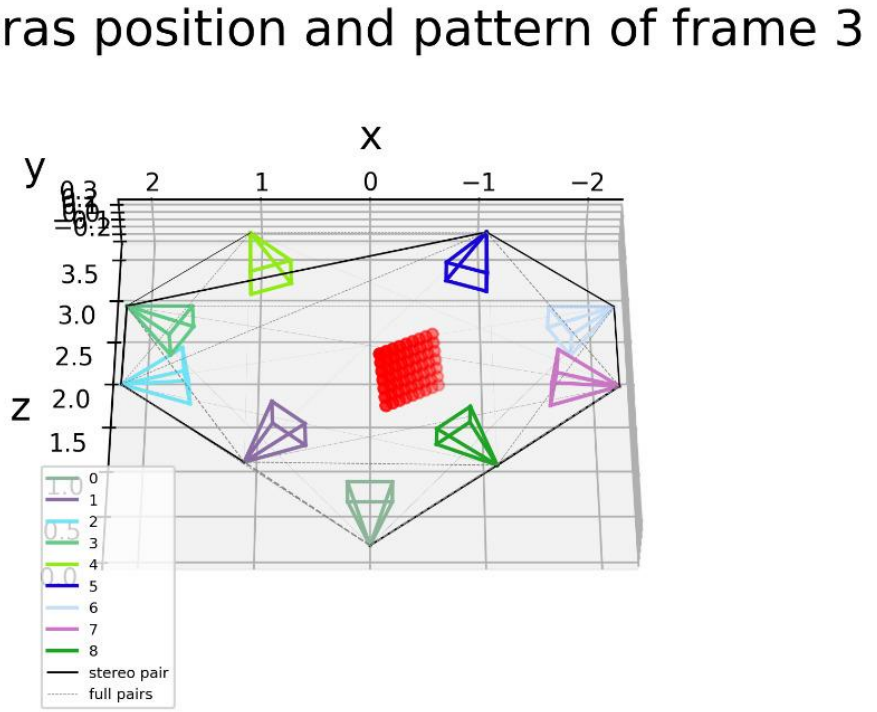
Per-camera intrinsics

Joint camera-pose refinement

Calibration with partial observations

Cameras can see different parts of the same target





ChArUco identifies the visible points

Overlap connects the camera rig

Synchronized views and a connected overlap graph are required.

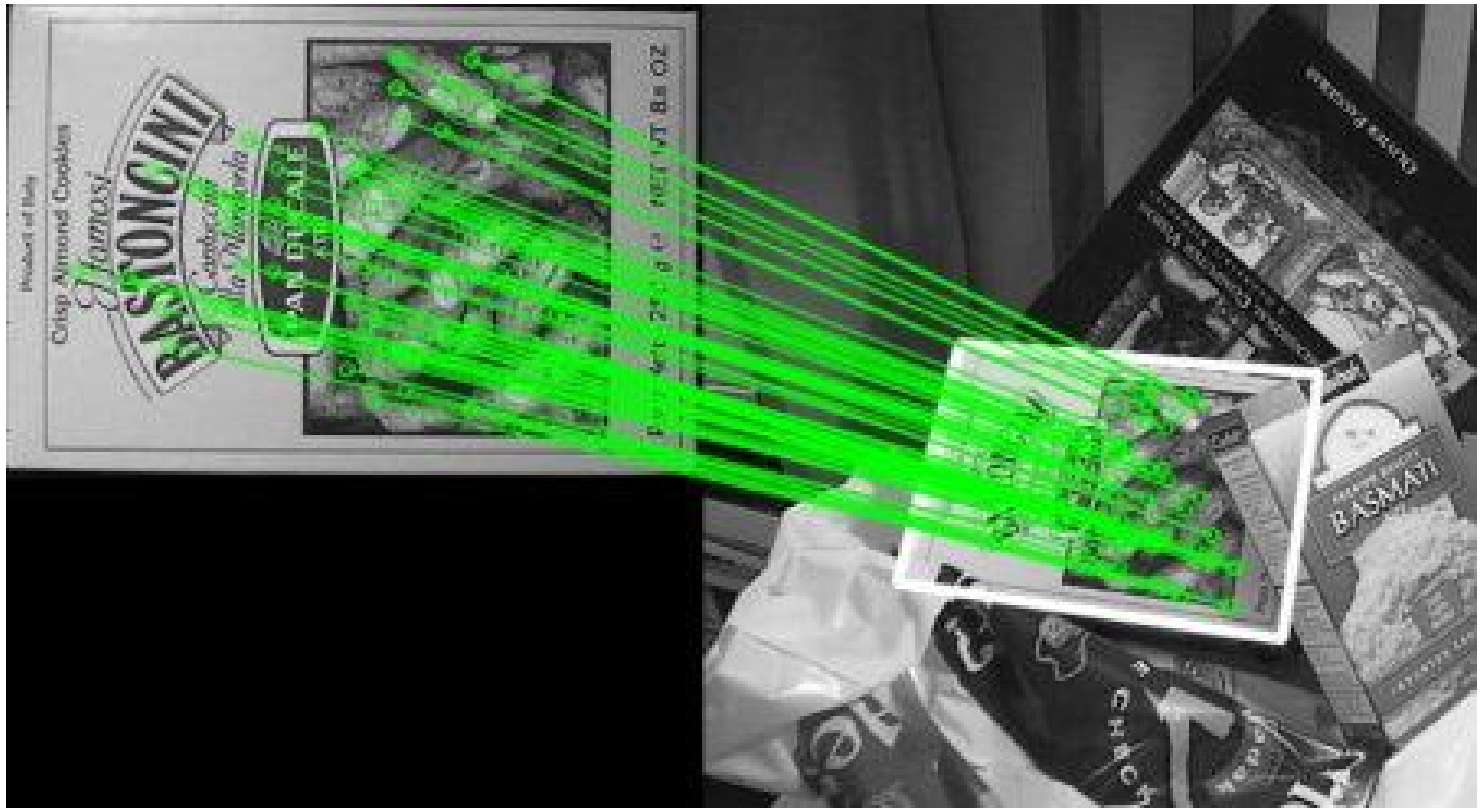
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Robust Geometry



Robust estimation with USAC



Geometry despite bad matches

Homography
Epipolar geometry
Camera pose

Better sampling and scoring
Local refinement
Early rejection

USAC also exists in OpenCV 4.x. Image illustrates feature matching.

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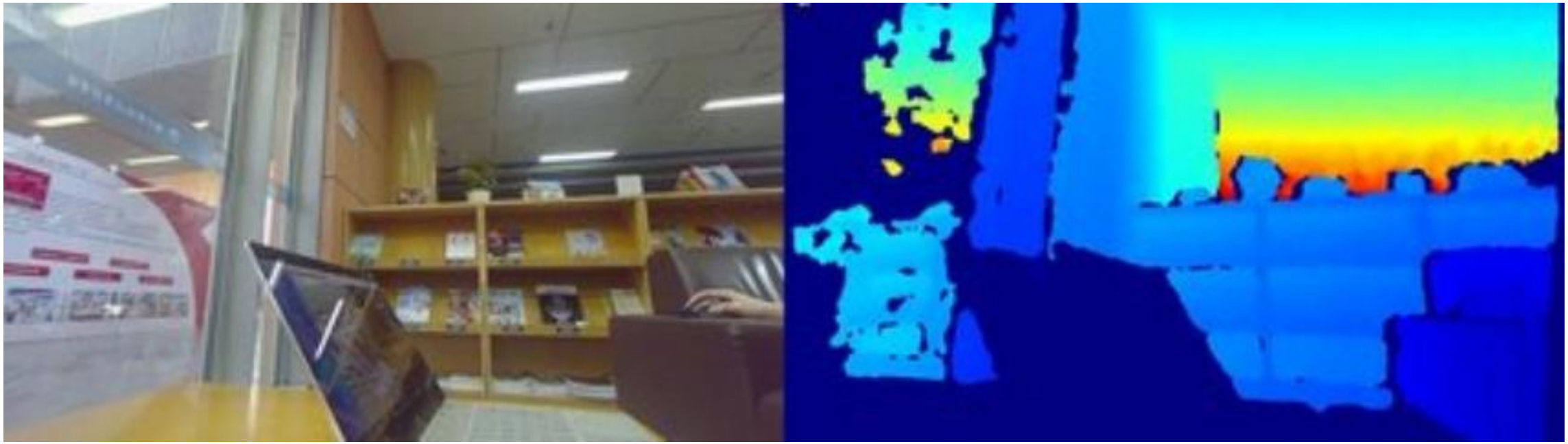
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3D Reconstruction



Depth fusion and camera motion

RGB-D input



Camera motion

Odometry and ICP alignment

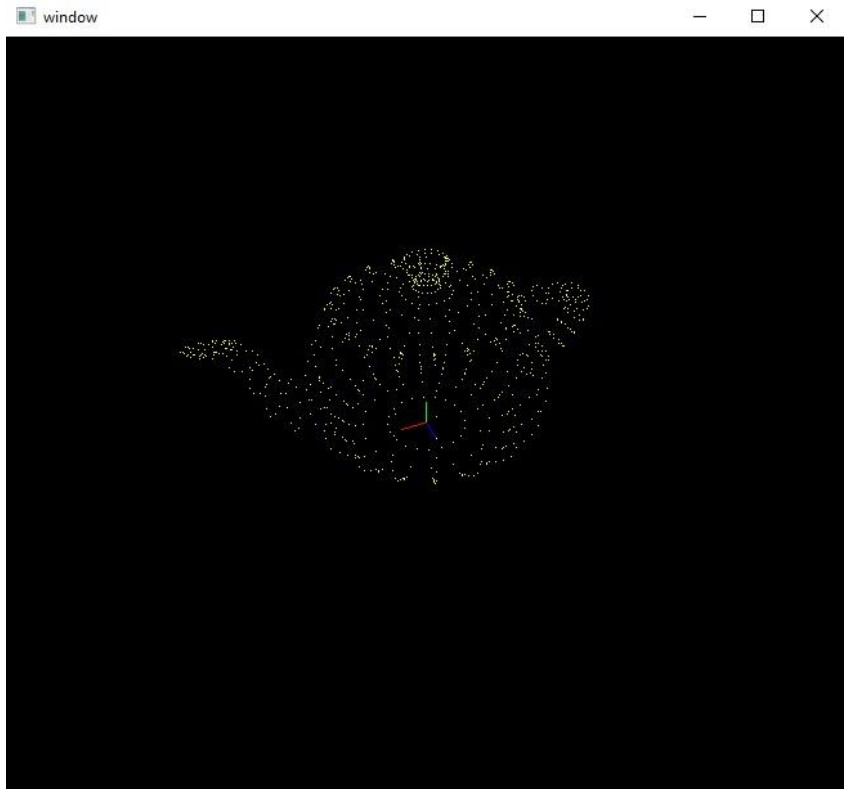
Surface reconstruction

Depth fusion with TSDF volumes

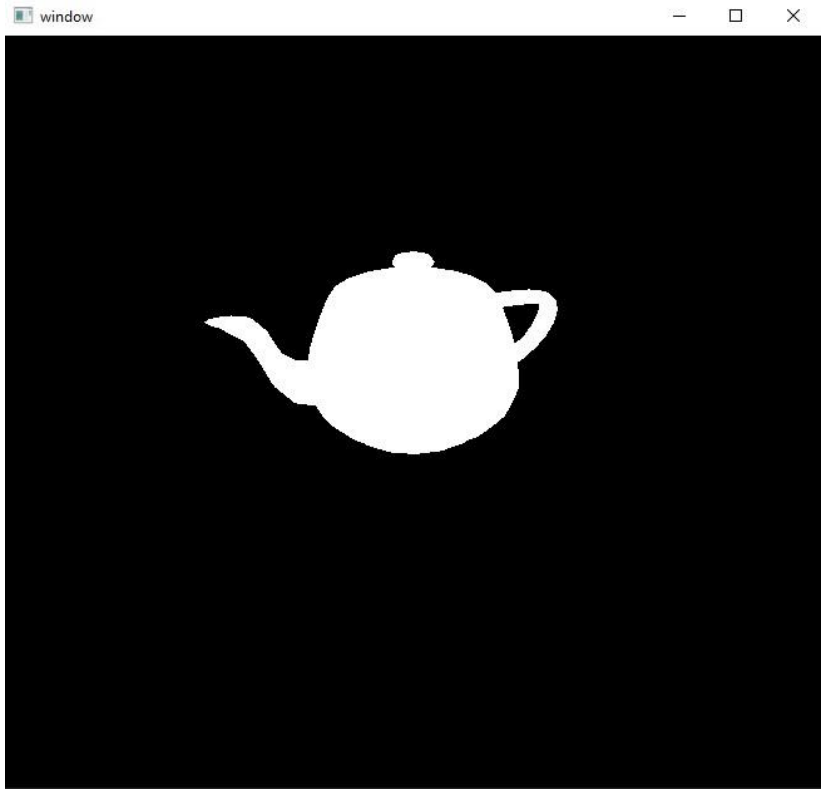
ptcloud brings tools from contrib/rgbd into the main library.

Point clouds and meshes

Point cloud



Triangle mesh



PLY / OBJ import and export

3D data tools in ptcloud

Official examples show two representations of a model.

Why upgrade to OpenCV 5

1. Better programming support
2. Performance New Hardware Acceleration Layer (HAL)
3. Modern Deep Neural Network (DNN) Engine
4. Better 3D vision support



Support for older OpenCV releases

5.x

Main development

1. New features
2. Security fixes
3. OpenCV Enterprise (paid) support available.

4.x

Maintenance

1. No new features
2. Security fixes until 2027)
3. OpenCV Enterprise (paid) support available

3.x / 2.x

End of Life (2023/2020)

1. No new features
2. No security fixes
3. OpenCV Enterprise (paid) support available



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OpenCV Enterprise



Three OpenCV Enterprise versions

LTS	Premier	Partnership
Stable-release maintenance	**Priority production support**	**Ongoing platform engineering**
Security backports Severe-defect fixes Certified builds	LTS services, plus Priority diagnosis More customer testing	Premier services, plus Reserved engineering Fork and hardware work
$150,000 / year	**$250,000 / year**	**From $400,000 / year**

All three include maintenance and OpenCV-certified builds for agreed configurations.

Indicative USD pricing for a three-year agreement, billed annually.
Initial qualification and taxes extra. Final pricing confirmed at launch.

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OpenCV 5
How to Upgrade



Upgrading to OpenCV 5

1	**Inventory**	APIs, dependencies, models and hardware
2	**Preserve a baseline**	Saved inputs and expected outputs
3	**Rebuild and verify**	Check correctness with OpenCV 5
4	**Benchmark and deploy**	Target hardware, gradual rollout and rollback



Testing the upgrade

Same representative inputs and target device

Baseline

Current OpenCV
Current model

Record expected outputs

Migration test

OpenCV 5
Same model

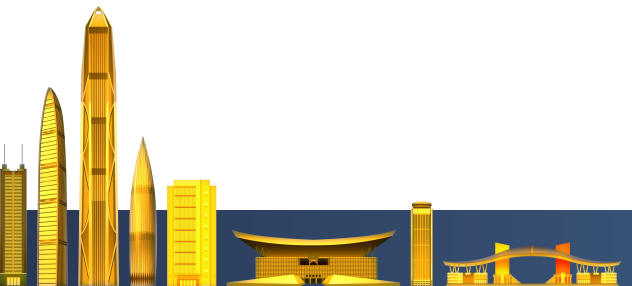
Isolate the runtime change

New-capability test

OpenCV 5
New model, if needed

Evaluate the model separately

Compare output quality, end-to-end latency and peak memory.
Measure the complete application, including time outside inference.

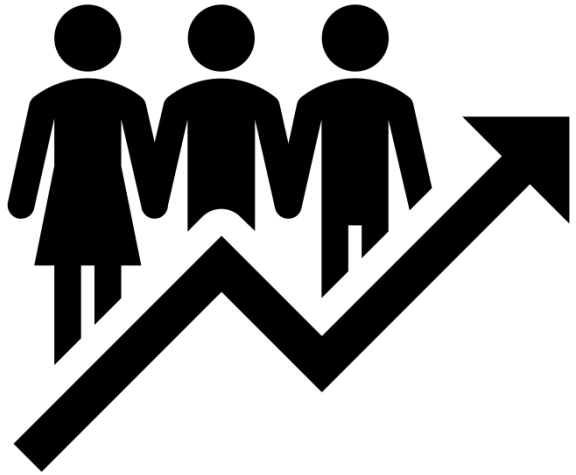


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OpenCV in China





A nonprofit team based in China supporting development of the OpenCV library

opencv.org.cn

OpenCV friends in China





Gold Members
(current and past)



Silver Member

OpenCV Enterprise Members in China

ORBBEC

3D camera access with OpenCV VideoCapture

RGB and depth streams

KHADAS

Quantized NPU inference with OpenCV DNN

Khadas VIM3 support since OpenCV 4.6 (June 2022)

Thank you!

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